Applied Microeconometrics (2026): Computer Assignment -1

Date: 7/8/2026

INSTRUCTIONS

Each assignment is in groups of two or three for the MA students and the PhD scholar will be working individually.

Choose a research paper that uses panel data but you should not replicate the paper as it is! You have to bring in some change either in terms of the nature of cross-section entities or the time-series or any new variable/ or their definition etc.

The number of cross section observations should be larger than the number of time-series observations.

Choose a limited number of explanatory variables out of which at least

(a) two variables are both entity-variant and time-variant; (b) one is time-invariant

(1) Read the paper and understand the analysis carefully.

(2) Please note the deadlines and a google link will be shared shortly to upload the files sequentially as indicated below.

- Please edit the and make the necessary changes in the team members.
- **19/8/2026**: Reference paper with the file name as **G#paper.pdf**.
- **29/8/2026**: Data file with name **G#.dta**(STATA file)**or G#.xls** (excel sheet if using any other software)..
- **2/9/2026**: The pdf file of the assignment **G#.pdf**.
- Note that # stands for the group number mentioned by you in the Assign teams file. Make any changes to the assignment teams if you wish to.
- Presentations will be in the week of 7th September and further details will be announced closer to the date.

(2) In the first page of the analysis file (pdf) mention the following:

- Names of team members
- The citation of the reference paper you are using for the assignment and
- Citations of any other web resources you have used to access program codes commands and any other information.

(3) Present your results in tables very neatly with proper table numbers and titles, column headings, appropriate * for significance. Marks will be deducted if STATA or any other software outputs are copy pasted as tables. Recall the commands of asdoc or outreg in STATA for last semester or estimates table as given in Torres Reyna's STATA to organize the tables. Do not present all the results displayed by the software. Refer to the reference paper and the textbook on how the results should be presented.

(4) The econometric equation for each of the model type has to be correctly specified as you read in textbook or mentioned in the sessions. Sometimes the reference paper may have some minor errors, and you have to be careful not to repeat that while specifying the equations.

(5) For the test of hypotheses, state clearly, the null and alternative hypotheses, along with the test statistics, degrees of freedom, and the decision.

QUESTIONS

Section 1

(1) State what is the objective of the paper and how does panel data help in analyzing the objective of the study in a better manner.
(2) Discuss how your analysis is different from the reference paper and link it to the objective of the reference paper. (e.g.) if the reference paper is using net FDI as one of the variables then you can change it to FDI/GDP that is, ratio of net FDI inflow to GDP. Then explain what is the relevance of this variable to the objective of the analysis.
(3) Given the above objective, what are the dependent and independent variables in the model. Make sure that you also clearly indicate whether a log transformation or a quadratic specification is important for the analysis. Sometimes you can use lag of one or two independent variables as additional explanatory variables.
(4) Mention the data source(s) and variables used with their proper definition. (e.g.) if you are using net FDI as one of the variables then define the variable as it is mentioned in the data source. Make sure the constant price values or PPP adjusted values are all properly taken into account.
(5) Mention clearly the number of cross-section entities and any specific features they have (e.g. 100 low and middle income countries) and the number of years with the years mentioned (e.g. 3 years 2005, 2010 and 2015). Make sure that N>=50 and N>>T.

Section 2

(6) Present a couple of graphical representation of the nature of variation of the data as may be interesting. Marks will be deducted if a cluttered graph is presented.
(7) Use the xtsum command of STATA or appropriate commands in the other software to present the summary statistics. Discuss the findings from the summary statistics by taking the help of STATA manual or any other resource for other software.

Section 3.1

(8) Coefficient Estimates: Based on the objective of your analysis compare the results for the following models with and without time dummies
    a. Pooled Model
    b. FE model
    c. RE model
    d. For each of the models in a, b, c above present F-test results on whether year dummies are jointly significant or not.
    e. Discuss briefly the overall findings of the results in terms of coefficient significance, signs, magnitudes, differences in results across the models.
    f. Interpret the coefficient of the variable that is most relevant to the objective of the study.

Section 3.2

(9) Tests of Hypotheses: In all the model comparisons below, present the results along with the appropriate test of hypothesis for the best fitting model with and without time dummies.
    a. Pooled with FE
    b. Pooled with RE
    c. FE with RE
    d. Comment on the final results by linking it to the objective of the study.